Exercises

1. Evaluate using a number line.

a. 8 + 5

b. (−3) + 7

c. 2 + (−8)

d. (−4) + (−6)

e. 4 + (−4)

2. An elevator services the floors shown in the diagram.



Elevator buttons

a. The elevator is currently on the 3rd floor. The addition statement, (−4) + 3 + 5 + (−6) + (−3), represents the movement of the elevator. Describe the sequence of buttons that were pressed. What floor is the elevator on now?

b. The elevator starts on the ground floor and travels to the following floors: 4, 7, G, 3, and P1. Write an addition statement to show where the elevator is with respect to the starting position. What is the sum?

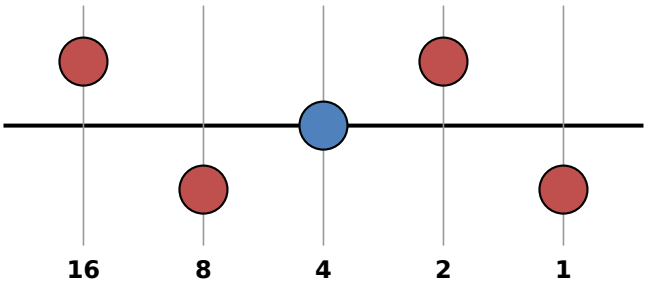
c. Describe a series of movements where the elevator travels a total of 15 floors.

3. If two different integers have a sum of zero, what special characteristic must be true about the two integers?

4. **a.** A binary abacus has 5 columns with the following features:

- The numbers 1, 2, 4, 8, and 16 represent the value of each column from right to left.
- The value of each column is negative if the bead is below the line, positive if the bead is above the line, and 0 if the bead is on the line.

The sum of the five columns represents a number. What number is being displayed?



b. Using the same 5 columns, place the beads so the abacus displays the number −11. Is there more than one way you can do this?

5. The Fibonacci numbers are the positive integers in the list 1, 1, 2, 3, 5, … Subsequent terms in the list are formed by adding the previous two numbers in the list. Thus, 1 + 1 = 2, 1 + 2 = 3, 2 + 3 = 5, …

a. Using the defining rule for the Fibonacci numbers, determine the next seven terms in this list.

b. If the initial two numbers were (−1) and (−1), what would the next "negative" Fibonacci number be?

c. Show the first 10 numbers in the list of "negative" Fibonacci numbers using the same rule from part a.

d. What do you observe about the list of "negative" Fibonacci numbers relative to the usual Fibonacci numbers?

6. To determine the sign of a sum of two integers, it sometimes is enough to just examine the signs of the individual terms of the sum. For example, as the table below shows, if two integers are both positive, then their sum is also positive.

Sign of Integer 1	Sign of Integer 2	Sign of Integer Sum
positive	positive	positive
positive	negative	
negative	positive	
negative	negative	

Complete the table. If the sign of the sum could be positive, negative, or zero (depending on the integers), label the sum in that case "either" and explain, in words, when each possible outcome occurs. Use examples to support your answers.

7. An online merchant who sells a popular game collects all orders for the game received during the day. Then, at midnight, she fills these orders for the game from her Starting Inventory.

a. Explain why the Starting Inventory and the Total Daily Orders for the game should each be integers.

b. To update her Inventory, the merchant finds the sum of the Starting Inventory and the **opposite** of today's Total Daily Orders. What does a positive value for the Updated Inventory mean?

c. As the table below indicates, on Day 1 the Updated Inventory is 15 + (−3) = 12, which becomes the Starting Inventory on Day 2. Then, on Day 2 the Updated Inventory is 12 + (−2) = 10. Compute the rest of the table in a similar fashion.

	Day 1	Day 2	Day 3	Day 4	Day 5

Starting Inventory	15	12	10		
Total Daily Orders	3	2	4	5	2
Updated Inventory	12	10			

d. What does the value of the Updated Inventory on Day 5 mean?

8. A frog starts at 0 on a number line. It jumps 100 units to the right, then jumps 200 units to the left. It then jumps 199 units to the right followed by 198 units to the left, and then 197 units to the right followed by 196 units to the left. The frog continues to move in alternate directions. Each movement is 1 unit less than the previous movement.

a. Where on the number line is the frog after 10 moves?

b. Where on the number line is the frog after 51 moves?

c. After how many jumps did the frog land on the number 50?

d. After how many jumps did the frog land on the number (−50)?

Source: CEMC Open Courseware, Grades 7 & 8, Lesson 1: Adding Integers (University of Waterloo).